

| Part Name / Description           | Qty | Unit Cost | Total Cost | Manufacturer                        | Manufacturer #      | Supplier         |
|-----------------------------------|-----|-----------|------------|-------------------------------------|---------------------|------------------|
| Barrel Jack Connector             | 1   | \$ 0.71   | \$ 0.71    | Same Sky                            | PJ-102AH            | DigiKey          |
| 3.3V Switching Voltage Regulator  | 1   | \$ 2.16   | \$ 2.16    | onsemi                              | LM2575D2T-3.3R4G    | DigiKey          |
| ESP32 Microcontroller             | 1   | \$ 5.06   | \$ 5.06    | Espressif Systems                   | ESP32-S3-WROOM-1-N4 | DigiKey, Peralta |
| OLED Screen                       | 1   | \$ -      | \$ -       | GoldenMorning Electronic            | GME12864-17         | Peralta          |
| 12mm Pushbutton                   | 7   | \$ 0.60   | \$ 4.20    | E-Switch                            | TL3300CF160Q        | DigiKey          |
| 3mm Pushbutton                    | 3   | \$ 0.90   | \$ 2.70    | Omron Electronics Inc-EMC Div       | B3U-1000P           | DigiKey          |
| USB Micro B Receptacle            | 1   | \$ 0.77   | \$ 0.77    | GCT                                 | USB3131-30-0230-A   | DigiKey, Peralta |
| Red LED                           | 1   | \$ 0.71   | \$ 0.71    | Visual Communications Company - VCC | CMD17-21VRD/TR8     | DigiKey          |
| Green LED                         | 1   | \$ 0.67   | \$ 0.67    | Visual Communications Company - VCC | CMD17-21VGD/TR8     | DigiKey          |
| Blue LED                          | 1   | \$ 0.45   | \$ 0.45    | QT Brightek (QTB)                   | QBLP653-IB5-2897    | DigiKey          |
| 100uF, 50V Electrolytic Capacitor | 1   | \$ 0.93   | \$ 0.93    | Panasonic Electronic Components     | EEE-FTH101XAP       | DigiKey          |
| 330uF, 25V Electrolytic Capacitor | 1   | \$ 0.85   | \$ 0.85    | Panasonic Electronic Components     | EEE-FK1E331P        | DigiKey          |
| 22uF, 25V Ceramic Capacitor       | 1   | \$ 0.33   | \$ 0.33    | Murata Electronics                  | GRM21BR61E226ME44K  | DigiKey          |
| 1uF, 25V Ceramic Capacitor        | 1   | \$ 0.12   | \$ 0.12    | KEMET                               | C0805C105K3RACTU    | DigiKey          |
| 0.1uF, 50V Ceramic Capacitor      | 13  | \$ 0.08   | \$ 1.04    | KEMET                               | C0805C104K5RACTU    | DigiKey          |
| 100, 1/4W Resistor                | 1   | \$ 0.10   | \$ 0.10    | YAGEO                               | RC1206JR-10100RL    | DigiKey          |
| 330, 1/4W Resistor                | 2   | \$ 0.11   | \$ 0.22    | YAGEO                               | RC1206FR-07330RL    | DigiKey          |
| 10k, 1/4W Resistor                | 10  | \$ 0.10   | \$ 1.00    | Stackpole Electronics Inc           | RMCF1206JG10K0      | DigiKey          |
| 220uH, 1.38A, 380mOhm Inductor    | 1   | \$ 1.01   | \$ 1.01    | Bourns Inc.                         | SRR1260-221K        | DigiKey          |
| 40V, 3A Schottky Diode            | 1   | \$ 0.69   | \$ 0.69    | onsemi                              | SS34                | DigiKey          |
| 800mA Fuse 5x20mm                 | 1   | \$ 0.57   | \$ 0.57    | SCHURTER Inc.                       | 34.1515             | DigiKey, Peralta |
| Cartridge Fuse Holder 5x20mm      | 1   | \$ 0.75   | \$ 0.75    | Keystone Electronics                | 4628                | Digikey, Peralta |
| 40 Pin Male Header Connector      | 1   | \$ -      | \$ -       | Lystaii                             | n/a                 | Peralta          |
| 40 Pin Female Header Connector    | 1   | \$ -      | \$ -       | Qunqi                               | n/a                 | Peralta          |
| 2-pin Female Jumper Connector     | 4   | \$ -      | \$ -       | n/a                                 | n/a                 | Peralta          |

| Vendor Link                                | Datasheet Link                             | Schematic Reference Designators                            | Sum Cost | Min Cost |
|--------------------------------------------|--------------------------------------------|------------------------------------------------------------|----------|----------|
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | J1                                                         | \$ 25.04 | \$ 18.50 |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | U1                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | U2                                                         |          |          |
| n/a                                        | [Link](https://www.ti.com/lit/pdf/slvu001) | U12                                                        |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | U5, U6, U7, U8, U9, U10, U11                               |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | U3, U4, U13                                                |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | J5                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | D4                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | D2                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | D3                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | C1                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | C2                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | C4                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | C6                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | C3, C5, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17 |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | R11                                                        |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | R1, R13                                                    |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | R2, R3, R4, R5, R6, R7, R8, R9, R10, R12                   |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | L1                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | D1                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | F1                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | [Link](https://www.ti.com/lit/pdf/slvu001) | F1                                                         |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | n/a                                        | J2, J3, J4, J6, J7, J8, J9, J10, J11, J12                  |          |          |
| [Link](https://www.ti.com/lit/pdf/slvu001) | n/a                                        | U12                                                        |          |          |
| n/a                                        | n/a                                        | J3, J4, J6, J7, J9, J11                                    |          |          |